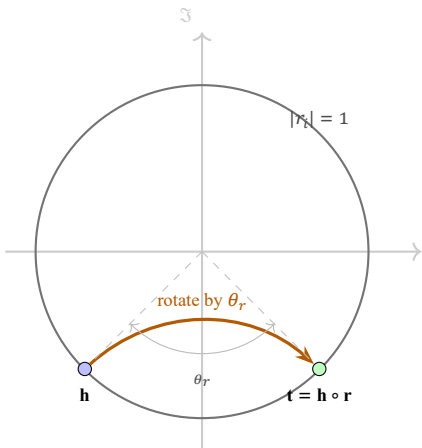




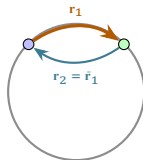
**Four relational patterns — one rotation mechanism:**

**Symmetry:**  $\theta_r \in \{0, \pi\} \Rightarrow r_i = \pm 1$

**Antisymmetry:**  $\theta_r \notin \{0, \pi\}$

**Inversion:**  $r_2 = \bar{r}_1, \theta_{r_2} = -\theta_{r_1}$

**Composition:**  $r_3 = r_1 \circ r_2, \theta_{r_3} = \theta_{r_1} + \theta_{r_2}$



reverse rotation